

LACOMBE, AB, Canada

WMO: 712420

Lat: 52.4492N

Lon: 113.7569W

Elev: 860

StdP: 91.41

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.6	-28.0	-34.7	0.2	-31.6	-31.1	0.2	-27.9	10.3	-4.2	9.0	-7.5	1.4	170	0.659

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	28.1	18.0	26.2	17.2	24.5	16.3	19.6	26.0	18.3	24.6	17.2	23.1	3.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	13.6	23.2	15.9	12.5	21.7	14.7	11.6	20.4	59.8	26.2	55.5	24.7	51.7	23.1	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
9.3	8.0	6.8	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-36.3	31.5	3.9	1.9	-39.1	32.8	-41.4	33.9	-43.6	35.0	-46.5	36.4
			-36.3	21.7	3.8	1.5	-39.0	22.8	-41.2	23.6	-43.4	24.5	-46.1	25.5
				DB										
				WB										

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.8	-11.1	-9.6	-4.3	3.6	9.8	13.7	16.3	15.1	10.4	3.9	-4.4	-10.2
	DBStd	11.49	8.61	7.62	7.69	5.13	4.42	3.03	2.91	3.15	4.19	4.63	6.82	7.55
	HDD10.0	3206	655	548	442	198	58	4	1	2	44	194	432	627
	HDD18.3	5681	913	782	700	443	266	141	77	107	238	448	682	885
	CDD10.0	591	0	0	0	4	52	117	195	161	57	5	1	0
	CDD18.3	24	0	0	0	0	1	3	13	7	1	0	0	0
	CDH23.3	636	0	0	0	2	49	72	239	201	69	3	0	0
	CDH26.7	126	0	0	0	0	7	9	53	43	14	0	0	0
Wind	WSAvg	2.8	2.4	2.4	2.7	3.4	3.5	3.2	2.8	2.5	2.7	2.9	2.5	2.4
Precipitation	PrecAvg	454	20	17	16	22	49	77	88	69	51	18	15	18
	PrecMax	566	67	44	40	56	136	186	186	168	118	49	35	54
	PrecMin	307	1	0	0	0	10	26	15	10	3	1	1	1
	PrecStd	70	15	11	9	16	28	40	40	39	30	13	10	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	7.7	13.4	22.4	27.6	27.8	30.5	30.6	28.7	22.4	14.0	7.5
		MCWB	3.2	3.2	6.4	11.1	14.1	17.3	20.0	18.5	16.3	12.0	7.9	3.2
	2%	DB	5.1	5.1	9.9	19.2	24.6	25.4	28.2	27.5	25.3	17.9	9.4	4.7
		MCWB	1.9	1.6	4.6	9.7	13.1	16.0	19.4	17.9	15.5	10.4	4.8	1.5
	5%	DB	2.7	3.0	7.6	15.8	22.0	23.4	26.0	25.6	22.6	14.8	6.3	2.3
		MCWB	0.3	0.1	3.4	7.6	12.1	15.0	18.5	17.4	14.1	8.5	2.8	-0.2
	10%	DB	0.3	0.8	5.4	12.8	19.4	21.3	24.0	23.5	19.7	12.1	4.1	0.1
		MCWB	-1.5	-1.3	2.0	6.0	10.9	14.0	17.3	16.7	12.8	6.9	1.4	-1.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	3.5	7.1	11.6	15.3	18.9	21.9	20.5	17.4	12.8	8.0	3.7
		MCDB	6.8	7.2	12.8	21.3	24.8	25.7	27.2	26.7	26.3	20.8	13.6	7.1
	2%	WB	2.1	1.8	5.0	9.9	13.7	17.1	20.2	19.1	15.7	10.7	5.1	1.6
		MCDB	4.9	4.9	9.3	18.6	22.7	23.3	26.5	25.6	23.7	17.3	8.9	4.3
	5%	WB	0.5	0.2	3.5	8.1	12.6	15.7	18.9	18.0	14.5	8.9	3.2	-0.1
		MCDB	2.6	2.7	7.2	15.0	20.8	21.7	25.1	24.4	21.9	14.0	5.9	2.1
	10%	WB	-1.5	-1.3	2.1	6.4	11.4	14.6	17.7	16.8	13.0	7.2	1.5	-1.8
		MCDB	0.3	0.7	5.1	12.0	18.5	20.3	23.3	22.8	19.0	11.4	3.7	0.1
Mean Daily Temperature Range	5% DB	MDBR	11.2	12.6	11.9	12.6	13.4	12.0	12.8	13.8	13.9	12.8	10.6	10.7
		MCDBR	13.0	14.7	14.5	18.8	18.4	16.2	16.5	18.4	20.2	17.8	13.7	12.0
		MCWBR	10.3	11.3	9.8	10.3	8.8	8.0	8.8	9.8	10.8	10.8	9.5	9.3
	5% WB	MCDBR	12.8	14.2	13.9	17.6	16.6	14.1	15.4	16.5	18.8	17.3	13.1	11.8
		MCWBR	10.2	11.0	9.6	10.1	8.1	7.4	8.6	9.3	10.3	10.6	9.3	9.3
Clear-Sky Solar Irradiance	taub	0.261	0.266	0.294	0.325	0.337	0.344	0.346	0.358	0.309	0.283	0.254	0.244	
	taud	2.312	2.390	2.409	2.376	2.399	2.402	2.422	2.362	2.482	2.536	2.470	2.319	
	Ebn at Noon	748	868	906	912	914	908	898	866	880	842	772	706	
	Edh at Noon	60	75	91	107	110	111	107	108	84	65	51	49	
All-Sky Solar Radiation	RadAvg	1.06	2.08	3.48	4.68	5.66	5.99	6.30	5.17	3.60	2.15	1.11	0.79	
	RadStd	0.09	0.17	0.24	0.42	0.37	0.40	0.34	0.35	0.39	0.24	0.11	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (76 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page